

2.4 Nitrogen & Sulfur

Question Paper

Course	CIEA Level Chemistry
Section	2. Inorganic Chemistry
Topic	2.4 Nitrogen & Sulfur
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

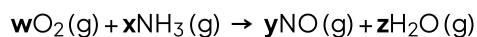
Which gas is present in the largest proportion in exhaust fumes of a car engine?

- A. Carbon monoxide
- B. Carbon dioxide
- C. Watervapour
- D. Nitrogen

[1 mark]

Question 2

Oxidation of ammonia by oxygen is the first step in the manufacture of nitric acid, which is used in the production of the synthetic material nylon.



Which values for **w**, **x**, **y** and **z** balance this equation?

	w	x	y	z
A	5	4	4	6
B	6	4	4	5
C	6	5	5	4
D	5	6	6	4

[1 mark]

Question 3

Using your knowledge of the Bronsted-Lowry theory, which of the following correctly describes ammonia?

- A. Neutral
- B. Acid
- C. Base
- D. Amphoteric

[1 mark]

Question 4

Which statement correctly describes an ammonium ion, NH_4^+ ?

- A. The bonds are all the same length.
- B. All bond angles are 107°
- C. Ammonium ions are unreactive when heated with NaOH (aq) .
- D. Ammonium ions are formed when ammonia behaves as an acid.

[1 mark]

Question 5

Nitrogen dioxide NO_2 , sulfur dioxide SO_2 , and carbon monoxide CO are all atmospheric pollutants.

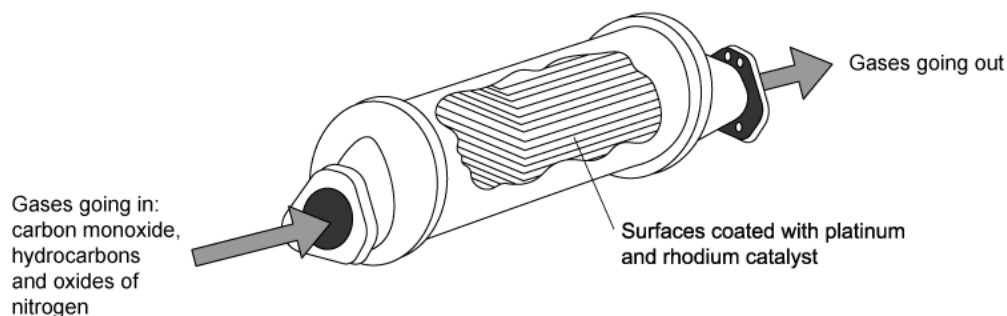
Which of the following reactions takes place in the atmosphere?

- A. NO_2 is reduced to NO by CO
- B. NO_2 is reduced to NO by SO_2
- C. SO_2 is oxidised to SO_3 by CO_2
- D. CO is spontaneously oxidised to CO_2

[1 mark]

Question 6

Which reaction that takes place in a catalytic converter does **not** remove hazardous and polluting gases from a car exhaust fumes?



These equations are qualitative and unbalanced.
[HC = unburnt hydrocarbons; NO_x = oxides of nitrogen]

- A. $\text{CO} + \text{O}_2 \rightarrow \text{CO}_2$
- B. $\text{CO} + \text{NO}_x \rightarrow \text{CO}_2 + \text{N}_2$
- C. $\text{HC} + \text{NO}_x \rightarrow \text{H}_2\text{O} + \text{CO}_2 + \text{N}_2$
- D. $\text{HC} + \text{NO}_x \rightarrow \text{H}_2\text{O} + \text{CO} + \text{N}_2$

[1 mark]

Question 7

Phosphorus and nitrogen are both Group V elements. Phosphorus forms a chloride with the formula PCl_5 .

Why is it **not** possible for nitrogen to form NCl_5 ?

- 1 Nitrogen is almost inert.
 - 2 Nitrogen cannot have oxidation state +5.
 - 3 Nitrogen's outer shell can only contain eight electrons.
- A. 1 only
 - B. 1 and 2
 - C. 3 only
 - D. 1, 2 and 3

[1 mark]

Question 8

The manufacture of ammonium nitrate NH_4NO_3 , occurs on a large scale for use in fertiliser.

Which statement about ammonium nitrate fertiliser is **not** correct?

- A. It is insoluble in water.
- B. It consists of 35 % nitrogen by mass.
- C. It can cause environmental problems.
- D. Nitric acid is used in its manufacture.

[1 mark]

Question 9

Oxides of nitrogen are released into the atmosphere in car engine exhaust fumes.

Which environmental problem is not made worse by this occurrence?

- A. Photochemical smog
- B. The hole in the ozone layer
- C. Corrosion of buildings
- D. Acidification of lakes

[1 mark]

Question 10

Which of the following transformations is sulfur dioxide involved in, as either a reagent or a catalyst, under atmospheric conditions?

- 1 NO_2 to NO
- 2 NO to NO_2
- 3 CO to CO_2

- A. 1 only
- B. 1 and 2
- C. 2 and 3
- D. 1, 2 and 3

[1 mark]

Model Answers: Easy

1

The correct answer is **D** because:

- **Air** is taken into the engine to provide oxygen for fuel combustion.
- **Nitrogen** gas makes up **79% of air**, therefore is present in the greatest amount when gas (exhaust fumes) leaves the engine.
- Although some nitrogen reacts with oxygen within the engine to produce nitrous oxides, these are removed by the **catalytic converter**.

A is incorrect as carbon monoxide is **removed** from the exhaust fumes by the catalytic converter.

B is incorrect as carbon dioxide only makes up **0.04% of air**, the amount of CO_2 produced by the engine will not be near the proportion of nitrogen.

C is incorrect as water vapour will not account for as much as nitrogen.

2

The correct answer is **A** because:

- List the number of each atom on either side of the unbalanced reaction:

Reactants	Products
N x 1	N x 1
H x 3	H x 2
O x 2	O x 2

- Look for differences in the equation, start with elements like metals and carbons, then oxygens and then hydrogens.

- There are **equal** numbers of O and N but **unequal** numbers of H on either side.
- Change the numbers to match by adjusting the coefficients.
 - $5\text{O}_2(\text{g}) + 4\text{NH}_3(\text{g}) \rightarrow 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{g})$

Reactants	Products
N x 4	N x 4
H x 12	H x 12
O x 10	O x 10

B is incorrect as this gives 10 reacting oxygen but 9 product oxygen and 12 reacting hydrogen but 10 product hydrogen.

C is incorrect as this gives 12 reacting oxygen but 9 product oxygen and 15 reacting hydrogen but 8 product hydrogen.

D is incorrect as this gives 18 reacting hydrogen but 8 product hydrogen.

3

The correct answer is **C** because:

- The **Bronsted-Lowry theory** describes acids as **proton donors** and bases as **proton acceptors**.
- Ammonia** acts as a Bronsted-Lowry **base** as it accepts a proton (H^+) to produce an **ammonium ion** (NH_4^+).
- Ammonium is formed by the nitrogen donating its **lone pair** to a hydrogen ion to form a **dative bond**.
- The reaction of ammonia and water is as follows:
 - $\text{NH}_3 + \text{H}_2\text{O} \rightarrow \text{NH}_4^+ + \text{OH}^-$

A is incorrect as ammonia is **not neutral** because it reacts in acid-base reactions due to its ability to accept a proton due to its lone pair.

B is incorrect as ammonia **cannot** act as a Bronsted-Lowry acid because it **cannot**

donate a proton.

D is incorrect as ammonia is **not amphoteric** (the ability to act as an acid and a base) because it does **not** act as an **acid**.

4

The correct answer is **A** because:

- All bonds are between nitrogen and hydrogen.
- Therefore the **distance** between the two **nuclei** of each bond is the **same**.
- The difference in **electronegativity** and **bond strength** is the same in all four N-H bonds and therefore the **bond length** is the **same**.

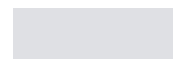
B is incorrect as the ammonium ion has a **tetrahedral shape** and therefore all bond angles are **109.5** (ammonia, NH_3 , has a lone pair of electrons and all bond angles are 107°).

C is incorrect as ammonium ions react with aqueous sodium hydroxide (NaOH) to produce ammonia (NH_3) and a sodium salt. Eg. $\text{NH}_4\text{Cl} + \text{NaOH} \rightarrow \text{NH}_3 + \text{NaCl} + \text{H}_2\text{O}$

D is incorrect as ammonium ions are formed when ammonia acts as a **base** and accepts a proton (H^+ ion).

5

The correct answer is **B** because:



- All bonds are between nitrogen and hydrogen.
- Therefore the **distance** between the two **nuclei** of each bond is the **same**.
- The difference in **electronegativity** and **bond strength** is the same in all four N-H bonds and therefore the **bond length** is the **same**.

B is incorrect as the ammonium ion has a **tetrahedral shape** and therefore all bond angles are **109.5** (ammonia, NH_3 , has a lone pair of electrons and all bond angles are 107°).

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D is incorrect as ammonium ions are formed when ammonia acts as a **base** and accepts a proton (H^+ ion).

5

The correct answer is **B** because:

- The following reaction occurs in the atmosphere during the production of acid rain:
 - $\text{NO}_2 + \text{SO}_2 \rightarrow \text{NO} + \text{SO}_3$
- In this reaction the oxidation numbers of the reactants are as follows:
 - NO_2 – Nitrogen (+4) Oxygen (-2), SO_2 – Sulfur (+4) Oxygen (-2)
- The oxidation numbers of the products are as follows:
 - NO – Nitrogen (+2) Oxygen (-2), SO_3 – Sulfur (+6) Oxygen (-2)
- Therefore, nitrogen has been **reduced** because the oxidation number is **lower**.

A is incorrect as carbon monoxide CO **does not** react with nitrogen dioxide NO_2 in the atmosphere.

C is incorrect as sulfur dioxide SO_2 is oxidised to sulfur trioxide SO_3 by **nitrogen dioxide** NO_2 not carbon dioxide.

D is incorrect as carbon monoxide is **not spontaneously oxidised** to carbon dioxide, this reaction takes place in a **catalytic converter** with a platinum catalyst.

6

The correct answer is **D** because:

- This reaction produces **carbon monoxide**, which causes suffocation as it binds irreversibly to hemoglobin (the protein in blood cells that carries oxygen).

A is incorrect as this reaction does occur in the **catalytic converter** to remove very dangerous carbon monoxide (CO) to produce CO₂ which is not directly hazardous to health but is a greenhouse gas.

B is incorrect as this is the overall reaction of the **catalytic converter**, removing toxic carbon monoxide and nitrous oxide to produce CO₂ and N₂ which are normal constituents of air.

C is incorrect as this reaction shows the **removal of unburnt hydrocarbons** (which cause smog that is harmful to inhale and causes visual impairment) to produce water vapour, CO₂ and N₂ which are normal constituents of air.

7

The correct answer is **C** because:

- Phosphorus can form PCl₅ because it can have **expanded octet**, this means it can **move** its lone pair electrons (3s orbital electrons) to **empty 3d orbital** during excited state and thus can form 5 bonds.
- Nitrogen **cannot** do this because the 2nd shell does **not** contain a d orbital, only 2s and 2p, and therefore cannot move any electrons to form more than 4 bonds (3 covalent and 1 dative).

A is incorrect as nitrogen is **fairly unreactive** because of the 3 covalent bonds between the two nitrogen atoms, but it is **not inert** and can react with chlorine to form NCl₃.

B & D are incorrect as nitrogen is not inert and can react with chlorine to form NCl₃, nitrogen can also have an **oxidation state** ranging from **-3** in ammonia and amines, to **+5** in nitric acid.

8

The correct answer is **A** because:

- Ammonium nitrate is an **ionic compound** that is **highly soluble** in water.

- In water NH_4NO_3 dissociates into an ammonium ion (NH_4^+) and a nitrate ion (NO_3^-).

B is incorrect as ammonium nitrate consists of 35% nitrogen by mass: Mass of N = $14 \times 2 = 28$. Mass of H = $1 \times 4 = 4$. Mass of O = $16 \times 3 = 48 \Rightarrow 28 \div (28 + 4 + 48) = 0.35 = \mathbf{35\%}$

C is incorrect as due to ammonium nitrate's solubility it is able to dissolve into water and produce a **slightly acidic solution** which is harmful to aquatic environments, ammonium nitrate is also very explosive.

D is incorrect as ammonium nitrate is produced when nitric acid reacts with ammonia : $\text{HNO}_3 + \text{NH}_3 \rightarrow \text{NH}_4\text{NO}_3$

9

The correct answer is **B** because:

- The formation of holes in the ozone layer is caused by chlorine containing organic molecules (haloalkanes).
- **Chlorine free radicals** are produced from haloalkanes under **UV light** and **catalyse** the depletion of ozone (O_3) to oxygen (O_2).

A is incorrect as nitrogen dioxide contributes to the **formation** of **ozone** and peroxyacetyl nitrate (**PAN**) which cause **photochemical smog**.

C is incorrect as nitrogen dioxide **catalyses** the formation of **SO_3** from SO_2 which results in the formation of acid rain – **acid rain corrodes buildings**.

D is incorrect as nitrogen dioxide **catalyses** the formation of **SO_3** from SO_2 which results in the formation of acid rain – **acid rain acidifies lakes**.

A is incorrect as nitrogen dioxide contributes to the **formation** of **ozone** and peroxyacetyl nitrate (**PAN**) which cause **photochemical smog**.

C is incorrect as nitrogen dioxide **catalyses** the formation of **SO₃** from SO₂ which results in the formation of acid rain – **acid rain corrodes buildings**.

D is incorrect as nitrogen dioxide **catalyses** the formation of **SO₃** from SO₂ which results in the formation of acid rain – **acid rain acidifies lakes**.

10

The correct answer is **A** because:

- During the production of acid rain sulfur dioxide (SO₂) reacts with nitrogen dioxide (NO₂) to produce sulfur trioxide (SO₃) and nitrogen monoxide (NO):
 - $\text{SO}_2(\text{g}) + \text{NO}_2(\text{g}) \rightarrow \text{SO}_3(\text{g}) + \text{NO}(\text{g})$
- SO₃ then reacts with water to produce **acid rain**:
 - $\text{SO}_3(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{SO}_4(\text{l})$

B is incorrect as sulfur dioxide is **not** involved in the production of NO₂ from NO, this is the reformation of NO₂ in its role as a **catalyst** in the acid rain reaction – $\text{NO}(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{NO}_2(\text{g})$.

C & D are incorrect as sulfur dioxide is **not** involved in the production of NO₂ from NO, this requires oxygen. Sulfur dioxide is **not** involved in the production of CO₂ from CO, this requires a **metal catalyst** and does not occur in atmospheric conditions (it occurs in the **catalytic converter**).

